SEPARATION OF SIMPLEST-FRACTION SETS IN A^2_1

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This proof note has evidence level Sketch: it has not appeared on arXiv or been peer-reviewed and should be checked by the reader. AI assistance was used in adapting and checking the argument and exposition.

> In particular, are pairwise distances between these sets bounded away from 0 in the space A^2_1?

Question 2, p. 9, of Evgeny Abakumov, Alexander Borichev and Konstantin Fedorovskiy, *Chui's conjecture in Bergman spaces*, arXiv:2009.01898, Math. Ann. 379 (2021), 1507–1532. Let SF_n consist of sums of n kernels $(z-a)^{-1}$ with $a \in \mathbb{T}$, counted with multiplicity, and equip A^2_1 with $\|f\|_1^2 = 2\int_{\mathbb{D}} |f(z)|^2(1-|z|^2)\,\mathrm{d}m_2(z)$, where $\mathrm{d}m_2 = \pi^{-1}\,\mathrm{d}x\,\mathrm{d}y$.

Answered: yes. Evidence: Sketch.

For all $n, k \geq 1$, $\operatorname{dist}_{A^2_1}(SF_n, SF_{n+k}) \geq \pi/\sqrt{6}$.

Proof. We encode the difference of the pole counts by a periodic discrepancy function. Its integer jumps force a uniform lower bound on the Fourier energy. Take $f = \sum_{j<n}(z-a_j)^{-1} \in SF_n$ and $g = \sum_{j<n+k}(z-b_j)^{-1} \in SF_{n+k}$, and write $a_j = e^{i\theta_j}$, $b_j = e^{i\psi_j}$. Expanding each kernel in the disc gives $f - g = -\sum_{m\geq 1} c_m z^{m-1}$, where $c_m = \sum_{j<n} a_j^{-m} - \sum_{j<n+k} b_j^{-m}$. Since the monomials are orthogonal in A^2_1 and $\|z^{m-1}\|_1^2 = 2/(m(m+1)) \geq 1/m^2$,

$$\|f-g\|_1^2 \geq \sum_{m\geq 1} |c_m|^2/m^2.$$

Define $F(t) = \#\{j : \theta_j \leq t\} - \#\{j : \psi_j \leq t\} + kt/(2\pi)$ on $[0, 2\pi]$. Its mth Fourier coefficient is $c_m/(2\pi i m)$. Parseval therefore identifies the last sum with $2\pi^2 \min_{c\in\mathbb{R}} \int_0^{2\pi} |F(t)-c|^2\,\mathrm{d}t/(2\pi)$. Because $F(t) - kt/(2\pi)$ is integer-valued, $|F(t)-c| \geq \operatorname{dist}(kt/(2\pi) - c, \mathbb{Z})$. Averaging over the integer number k of periods gives a squared mean of at least $1/12$. Hence $\|f-g\|_1^2 \geq \pi^2/6$, uniformly in the two functions and their indices. □

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